Exp:4 Best First Search

Program:

```
import heapq
def best_first_search(graph, heuristic, start, goal):
    priority_queue = [(heuristic[start], start)]
    visited = set()
    while priority_queue:
        h, node = heapq.heappop(priority_queue)
        if node in visited:
            continue
        print(node, end=" ")
        visited.add(node)
        if node == goal:
            print("\nGoal Reached!")
            return
        for neighbour in graph[node]:
            if neighbour not in visited:
                heapq.heappush(priority_queue, (heuristic[neighbour], neighbour))
    print("\nGoal Not Found!")
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['G'],
    'F': [],
    'G': []
}
heuristic = {
```

```
    'A': 6,
    'B': 4,
    'C': 5,
    'D': 3,
    'E': 2,
    'F': 4,
    'G': 0
}
best_first_search(graph, heuristic, 'A', 'G')
```

Output: